Binomial sign−conflict simulations (200 replicates)

Entries are mean (standard error) across replicates.

Model 7

blocks of 10, rho = 0.5, beta (1,1,1,−1) in blocks 1−3; n = 400, p = 1000; 3 sign conflicts

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	4.6286 (0.0922)	0.3502 (0.0088)	0.2942 (0.0057)	31.9600 (0.9299)	0.9780 (0.0008)	0.9133 (0.0068)	0.0213 (0.0009)
suni2D_lambda_alpha	4.6280 (0.0922)	0.3503 (0.0088)	0.2942 (0.0057)	31.9650 (0.9292)	0.9780 (0.0008)	0.9133 (0.0068)	0.0213 (0.0009)
suni2D_sum_rate	4.6280 (0.0922)	0.3503 (0.0088)	0.2942 (0.0057)	31.9700 (0.9287)	0.9779 (0.0008)	0.9133 (0.0068)	0.0213 (0.0009)
uniLasso	6.0361 (0.0370)	0.5337 (0.0107)	0.4421 (0.0086)	41.8800 (1.2324)	0.9641 (0.0012)	0.7492 (0.0010)	0.0333 (0.0012)
lasso	6.2367 (0.0734)	0.4471 (0.0099)	0.3857 (0.0056)	62.8850 (1.6626)	0.9465 (0.0016)	0.8925 (0.0064)	0.0528 (0.0016)
adalasso_uni	3.5539 (0.0637)	0.3808 (0.0099)	0.3335 (0.0074)	61.2300 (1.0727)	0.9500 (0.0011)	0.9700 (0.0033)	0.0502 (0.0011)
adalasso_lasso	4.1926 (0.0918)	0.4469 (0.0121)	0.4285 (0.0098)	31.1950 (0.6278)	0.9777 (0.0005)	0.8708 (0.0071)	0.0210 (0.0006)

Model 8

blocks of 10, rho = 0.7, beta (1,1,1,−1.5) in blocks 1−3; n = 400, p = 1000; 3 sign conflicts

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	5.6546 (0.1159)	0.3186 (0.0096)	0.2523 (0.0054)	40.1850 (0.7615)	0.9712 (0.0007)	0.9762 (0.0037)	0.0288 (0.0008)
suni2D_lambda_alpha	5.6546 (0.1159)	0.3186 (0.0096)	0.2523 (0.0054)	40.1850 (0.7615)	0.9712 (0.0007)	0.9762 (0.0037)	0.0288 (0.0008)
suni2D_sum_rate	5.6547 (0.1159)	0.3186 (0.0096)	0.2523 (0.0054)	40.1850 (0.7615)	0.9712 (0.0007)	0.9762 (0.0037)	0.0288 (0.0008)
uniLasso	11.2557 (0.0340)	0.6388 (0.0120)	0.4881 (0.0079)	34.6750 (1.0150)	0.9703 (0.0010)	0.7087 (0.0038)	0.0265 (0.0010)
lasso	9.9696 (0.1434)	0.5486 (0.0115)	0.4245 (0.0062)	64.1150 (2.1055)	0.9445 (0.0019)	0.8600 (0.0096)	0.0544 (0.0020)
adalasso_uni	4.2587 (0.0898)	0.3307 (0.0090)	0.2747 (0.0064)	59.0650 (0.8800)	0.9527 (0.0009)	0.9908 (0.0023)	0.0477 (0.0009)
adalasso_lasso	7.1291 (0.2206)	0.5019 (0.0137)	0.4720 (0.0106)	31.9400 (0.8736)	0.9755 (0.0007)	0.8113 (0.0112)	0.0225 (0.0008)

Model 9

blocks of 10, rho = 0.8, beta (2,2,−2) in blocks 1−3; n = 400, p = 1000; 3 sign conflicts

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	8.5272 (0.1877)	0.2335 (0.0082)	0.2075 (0.0048)	31.8950 (0.5842)	0.9771 (0.0006)	0.9994 (0.0006)	0.0231 (0.0006)
suni2D_lambda_alpha	8.5279 (0.1876)	0.2340 (0.0082)	0.2076 (0.0048)	31.9050 (0.5876)	0.9771 (0.0006)	0.9994 (0.0006)	0.0231 (0.0006)
suni2D_sum_rate	8.5272 (0.1877)	0.2335 (0.0082)	0.2075 (0.0048)	31.8950 (0.5842)	0.9771 (0.0006)	0.9994 (0.0006)	0.0231 (0.0006)
uniLasso	23.1502 (0.0797)	0.6988 (0.0157)	0.5746 (0.0098)	27.7000 (0.9494)	0.9753 (0.0009)	0.6650 (0.0010)	0.0219 (0.0010)
lasso	22.7570 (0.3277)	0.6386 (0.0144)	0.5407 (0.0079)	53.4300 (2.0793)	0.9519 (0.0019)	0.7956 (0.0106)	0.0467 (0.0020)
adalasso_uni	6.2200 (0.1487)	0.2627 (0.0088)	0.2284 (0.0059)	44.8950 (0.7513)	0.9641 (0.0008)	1.0000 (0.0000)	0.0362 (0.0008)
adalasso_lasso	15.4457 (0.5524)	0.5488 (0.0191)	0.4673 (0.0142)	20.8000 (0.5896)	0.9842 (0.0005)	0.7761 (0.0111)	0.0139 (0.0005)

Binomial sign−conflict simulations (200 replicates)

Entries are mean (standard error) across replicates.

Model 10

AR(1) rho = 0.8, beta (1,1,1,−1.5) at x1−4, x101−104, x201−204; n = 400, p = 1000; 3 sign conflicts

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	9.0194 (0.1516)	0.4298 (0.0114)	0.3422 (0.0064)	29.5750 (1.0279)	0.9761 (0.0008)	0.7404 (0.0116)	0.0209 (0.0009)
suni2D_lambda_alpha	9.0189 (0.1515)	0.4299 (0.0115)	0.3422 (0.0064)	29.5650 (1.0288)	0.9761 (0.0008)	0.7404 (0.0116)	0.0209 (0.0009)
suni2D_sum_rate	9.0189 (0.1515)	0.4297 (0.0114)	0.3422 (0.0064)	29.5700 (1.0283)	0.9761 (0.0008)	0.7404 (0.0116)	0.0209 (0.0009)
uniLasso	11.4973 (0.0372)	0.6071 (0.0126)	0.4895 (0.0086)	37.0600 (1.1891)	0.9638 (0.0012)	0.5363 (0.0043)	0.0310 (0.0012)
lasso	10.6988 (0.0914)	0.4988 (0.0113)	0.4099 (0.0057)	57.9900 (1.6833)	0.9463 (0.0015)	0.6804 (0.0081)	0.0504 (0.0016)
adalasso_uni	7.0953 (0.1470)	0.4356 (0.0121)	0.3707 (0.0083)	58.2350 (0.9688)	0.9507 (0.0009)	0.8804 (0.0077)	0.0482 (0.0009)
adalasso_lasso	9.3436 (0.1449)	0.4798 (0.0127)	0.4322 (0.0092)	25.7350 (0.6266)	0.9775 (0.0005)	0.6338 (0.0088)	0.0184 (0.0006)

Model 11

control without conflicts: blocks of 20, rho = 0.9, beta (1,1) in blocks 1−5; n = 100, p = 1000

	sse	rpe_class	rpe_kl	support	sign	tpr	fpr
suni2D_zeta_tau	11.7891 (0.2714)	0.6191 (0.0191)	0.5579 (0.0162)	17.5500 (0.2924)	0.9785 (0.0003)	0.3025 (0.0100)	0.0147 (0.0003)
suni2D_lambda_alpha	11.7817 (0.2710)	0.6192 (0.0192)	0.5582 (0.0162)	17.5350 (0.2928)	0.9785 (0.0003)	0.3030 (0.0100)	0.0147 (0.0003)
suni2D_sum_rate	11.8093 (0.2764)	0.6169 (0.0190)	0.5589 (0.0162)	17.5450 (0.2924)	0.9785 (0.0003)	0.3030 (0.0100)	0.0147 (0.0003)
uniLasso	13.4594 (0.3267)	0.6151 (0.0178)	0.5970 (0.0190)	20.6900 (0.3239)	0.9756 (0.0003)	0.3165 (0.0092)	0.0177 (0.0003)
lasso	10.8334 (0.1994)	0.6875 (0.0188)	0.6314 (0.0163)	28.1750 (0.4976)	0.9682 (0.0005)	0.3165 (0.0093)	0.0253 (0.0005)
adalasso_uni	13.5549 (0.3427)	0.6150 (0.0178)	0.5979 (0.0192)	20.7750 (0.3208)	0.9756 (0.0003)	0.3165 (0.0092)	0.0178 (0.0003)
adalasso_lasso	29.5864 (0.7144)	0.7048 (0.0204)	1.0748 (0.0344)	12.8650 (0.2150)	0.9809 (0.0003)	0.1900 (0.0078)	0.0111 (0.0002)